

Candidate Name: _____

Class Index No.

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Preliminary Examination 2008
Pre-University 3

MATHEMATICS
Higher 2

9740/01

MONDAY

8 SEPTEMBER

3 hours

Additional materials:

Cover Page

Writing papers

List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of **5** printed pages.

[Turn over]

1. Solve the simultaneous equations

$$z + (i - 1)w = 0,$$

$$3z + (1 - i)w = 1. \quad [4]$$

2. Expand $\frac{1}{(1-x)^2}$ in ascending powers of x up to and including the term in x^4 .

Deduce the term in x^n .

Hence by putting an appropriate value for x in your expansion, find the value of

$$\sum_{r=1}^{\infty} \frac{r}{2^{r-1}}. \quad [5]$$

3. The point A has coordinates (0,7,0) and the line L has equation $r = \lambda \begin{pmatrix} 1 \\ -5 \\ 3 \end{pmatrix}$ where

$\lambda \in \mathbb{R}$. Find the coordinates of the point B on L such that AB is perpendicular to L. [5]

4. The complex numbers v and w are given by $v = 2 - i$ and $w = -1 + 3i$.

Write down, in polar form, an expression for $\frac{v}{w}$.

Hence find the value of $\text{Im} \left[\left(\frac{v}{w} \right)^5 \right]$. [5]

5. Use mathematical induction to prove that for any positive integer n ,

$$\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4}.$$

Hence find an expression, in terms of n , for the sum of the series

$$2^3 + 4^3 + 6^3 + 8^3 + \dots + n^{\text{th}} \text{ term}.$$

Hence find the sum of the series $1^3 + 3^3 + 5^3 + \dots + 29^3$. [8]

6. (a) (i) Find the numerical value of the derivative of $\lg x^3$ when $x = 10$. [1]
- (ii) Hence find the equation of the normal to the graph of $y = \lg x^3$ at the point where $x = 10$, giving your answer in the form $y = mx + c$. [3]

(b) Show that $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx = 1$. Hence find the exact value of $\int_0^1 \sin^{-1} x \, dx$. [4]

7. (a) (i) State the value of $e^{i\pi}$. [1]
- (ii) Given that $z = e^{i\theta}$, solve for z if $2 - e^{i(\pi + \theta)} = i(z + 1)$. [2]

(b) Sketch, on a single clearly labelled Argand diagram, the loci defined by the equations

(i) $|z + 1| = 1$,

(ii) $\arg(z + 2) = \frac{\pi}{3}$.

Find the complex number w represented by the point of intersection of the two loci. [5]

8. (a) A sequence $\{x_n\}$ of positive numbers is defined as $x_{n+1} = \frac{1}{x_n^2 + 5}$ for $n \in \mathbb{Z}^+$.

The sequence $\{x_n\}$ converges to a number α as n tends to infinity.

Find the value of α . [3]

- (b) Given that $f(r) = \frac{1}{r^2}$ where r is a positive integer, express $f(r) - f(r + 1)$ as a single fraction.

Hence show that $\sum_{r=n}^{4n} \frac{2r+1}{r^2(r+1)^2} = \frac{(3n+1)(5n+1)}{n^2(4n+1)^2}$. [6]

9. (a) The curve $y = f(x)$ has gradient e^{2x} at the point (x, y) . Find $f(x)$ given that $y = 0$ when $x = \ln 3$. [3]

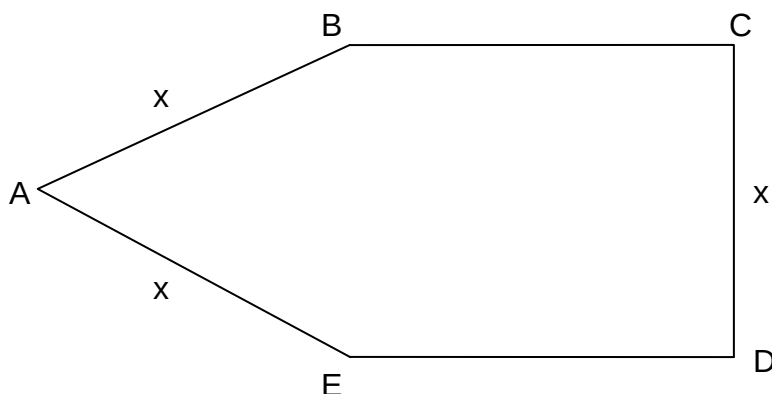
- (b) Show that the differential equation $xy \frac{dy}{dx} = x^2 + 2y^2$ may be reduced by means of the substitution $y = vx$ to $x \frac{dv}{dx} = \frac{1 + v^2}{v}$.

Hence obtain the general solution of y in the form $y^2 = F(x)$. [6]

10. (a) Express $f(x)$ in partial fractions where $f(x) = \frac{17 + x}{(4 - 3x)(1 + 2x)}$.

Given that, when $x = 0.25$, x increases at a constant rate of 1.5 unit/s, find the rate of change of $f(x)$ at this instant. [6]

- (b)



The diagram above shows a pentagon ABCDE of fixed perimeter P cm. Its shape is such that ABE is an equilateral triangle and BCDE is a rectangle. If the length of AB is x cm,

- (i) show that the area of ABCDE denoted by S is $\left(\frac{\sqrt{3}}{4} - \frac{3}{2}\right)x^2 + \frac{1}{2}Px$,
(ii) find the value of $\frac{P}{x}$ for which S is a maximum, leaving your answer in surd form. [6]

11. Two planes π_1 and π_2 have equations $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = 9$ and $\mathbf{r} \cdot \begin{pmatrix} 4 \\ 3 \\ -1 \end{pmatrix} = 8$ respectively.

The point P has coordinates (1,-1,3).

- (i) Show that P lies on plane π_1 . [2]
- (ii) Find the shortest distance from P to plane π_2 . [3]
- (iii) Find the acute angle between π_1 and π_2 . [3]
- (iv) Find, in scalar product form, an equation of the plane π_3 which passes through P and is perpendicular to π_1 and π_2 . [4]

12. Sketch, on a clearly labelled diagram, the graph of the curve $y = 1 + \frac{1}{4x^2+1}$. [2]

- (i) Shade on the diagram, a region A whose area is given by $\int_0^{1/2} \left(1 + \frac{1}{4x^2+1}\right) dx$ and show that the area of A is $\frac{1}{2} + \frac{\pi}{8}$ units². [3]
- (ii) By means of the substitution $2x = \tan\theta$, show that

$$\int_0^{1/2} \frac{1}{(4x^2+1)^2} dx = \frac{1}{8} + \frac{\pi}{16}. \quad [6]$$

- (iii) The region R is bounded by the curve, the lines $x = 0$, $x = \frac{1}{2}$ and the x-axis. Find the exact value of the volume of the solid formed when R is rotated completely about the x-axis. [4]